

Measurement of nuclear activation cross sections of protons on natural carbon for proton beam energies between 100 and 220 MeV

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ABSTRACT

In this work, we present measurements of nuclear activation cross sections from carbon. Carbon can be used for proton fluence calculation in proton therapy by measurements of the activity after irradiation. The cross sections are measured on a high-precision level with systematic uncertainties of approx. 2.8% for proton beam energies between 100 and 220 MeV. The targets are activated at the West German Proton Therapy Centre Essen. The determination of the activity is based on gamma-ray spectrometry measurements at the Dortmund Low Background Facility. For the $^{nat}\text{C}(p,x)^{11}\text{C}$ -reaction the results of this experimental study are comparable with several measurements reported in the literature. For the $^{nat}\text{C}(p,x)^7\text{Be}$ -reaction, the results calculated by several groups are reproduced by these measurements up to 180 MeV. Above 180 MeV, the observed energy dependence is weaker than the one measured by other groups. The reproducibility of the experimental setup is demonstrated with a measurement of the cross sections from aluminum.

1. Introduction

In proton therapy research the activation of several materials has been discussed for different applications over the last decades. On the one hand, the activation of graphite targets is suitable for proton fluence determination, which can be used for dose estimation [1]. On the other hand, positron emission tomography (PET) has been evaluated for proton beam range verification [2]. The first method uses the activation of natural carbon to measure the ^{11}C activity after the irradiation of the target. The second method compares a PET scan with a Monte Carlo (MC) simulation of the expected β^+ activation. Both methods require an accurate knowledge of the activation cross section.

The activation cross sections for the $^{nat}\text{C}(p,x)^{11}\text{C}$ -reaction have been measured as a function of the proton energy during the last seven decades by several groups [3–9] and most of the data have been collected in the EXFOR database [10]. Interestingly, the data points accumulate around two different excitation functions which are about

10% apart. These measurements mainly differ in two aspects: first, different half-lives have been used for the decay correction between irradiation and measurement. Second, some of the measurements have been normalized to a reference activation cross section.

These two aspects motivate a new measurement of the $^{nat}\text{C}(p,x)^{11}\text{C}$ -activation cross section with a focus on reducing uncertainties and using the latest half-life estimate. In order to determine the activation cross section, the target is irradiated at the West German Proton Therapy Centre (WPE). Subsequently, the activity of the generated radionuclides is measured at the Dortmund Low Background Facility (DLB) [11], a low-level gamma-ray spectrometry laboratory at TU Dortmund University.

2. Materials and methods

We have demonstrated in Ref. [12] that our setup is associated with small experimental uncertainties. While a detailed discussion of the

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experimental setup and its systematic uncertainties can be found in Ref. [12], we here discuss the measurement of nuclear activation cross sections from carbon and we compare the results with the data taken from the EXFOR database. We also investigate the reproducibility and stability over time by the irradiation of an aluminum sample with 100 MeV proton beam energy and a subsequent activation measurement.

2.1. Target materials

High-purity graphite sheets are used as target material. These sheets have a thickness of (1.05 ± 0.05) mm in beam direction, so the energy loss per unit length is varying to a negligible degree at this energy. The target area perpendicular to the beam direction is about $2.5 \text{ cm} \times 2.5 \text{ cm}$ and is motivated by the size of the proton field. The mass ranges between $(0.59 \pm 0.01) \text{ g}$ and $(0.61 \pm 0.01) \text{ g}$ for the different sheets used and is given by the different sizes of the target area. The uncertainty is limited by that of the digital scale. A new sample is used for each irradiation and measurement procedure. In order to maintain reproducibility of the experimental setup, a new aluminum target with the same dimensions as used in Ref. [12] is investigated. For both target materials the expected radionuclides available from Ref. [10] are summarized in Table 1. Since the target is brought from Essen to Dortmund, the observation of the reaction to ^{27}Mg is sensitive on the time for the transport which takes 40 min compared with the half-life of $(9.458 \pm 0.012) \text{ min}$ [13].

2.2. Target irradiation

The target is irradiated in one of the clinical treatment rooms at the WPE. The number of protons per Monitor Unit (MU) is measured with a Faraday-Cup (FC). The FC has an entrance window made of 0.7 mm thick stainless steel followed by the guard ring (160 mm length) and the collecting tube (120 mm length) made of aluminum. The collecting block has a diameter of 120 mm and is made of copper. Before the measurement, the FC is evacuated to about $1 \times 10^{-5} \text{ mbar}$. The guard potential is chosen as -1.2 kV according to Ref. [14]. The uncertainty of the FC measurements is dominated by the detection efficiency of the FC. Furthermore, the reproducibility is affected by output fluctuations of the treatment machine. Secondary particles as well as variations in the uniformity in the proton field have been identified as contributors to the uncertainty, leading to an overall estimate of 1% [12]. The FC has been placed in the isocenter about 50 cm away from the nozzle. Electrons produced in the output window can not reach the FC as well as the one from the entrance window of the FC are suppressed by the guard potential. The pencil beam scanning (PBS) mode is used for the activation of the target. The field size is $5.25 \text{ cm} \times 5.25 \text{ cm}$ full-width at half maximum, resulting in a homogenous proton fluence on the target area.

2.3. Measurement of the activity

The measurement of the activity of the produced radionuclides after irradiation is performed at the DLB. It is located above ground and has

Table 1

Summary of expected and obtained nuclear reactions with the half-life of the produced radionuclides from the literature [13].

Graphite		Aluminum	
Reaction	$T_{1/2}^{\text{lit}}$	Reaction	$T_{1/2}^{\text{lit}}$
$^{12}\text{C}(\text{p},\text{x})^{11}\text{C}$	$(20.361 \pm 0.023) \text{ min}$	$^{27}\text{Al}(\text{p},\text{x})^7\text{Be}$	$(53.22 \pm 0.06) \text{ d}$
$^{12}\text{C}(\text{p},\text{x})^7\text{Be}$	$(53.22 \pm 0.06) \text{ d}$	$^{27}\text{Al}(\text{p},\text{x})^{22}\text{Na}$	$(2.6029 \pm 0.0008) \text{ a}$
		$^{27}\text{Al}(\text{p},\text{x})^{24}\text{Na}$	$(14.958 \pm 0.002) \text{ h}$
		$^{27}\text{Al}(\text{p},\text{x})^{27}\text{Mg}$	$(9.458 \pm 0.012) \text{ min}$

an outer shielding consisting of barite concrete and cast iron, which corresponds to about 10 m water equivalent. The inner shielding consists of lead, copper and boron-loaded polyethylene and reduces the environmental radiation inside the DLB further. Since half of the cosmic muons flux is stopped in the massive outer shielding, the DLB features an active muon veto [15]. This suppresses the flux of cosmic muons by a factor of 12. Overall, both passive and active shielding techniques reduce the background by a factor of 4000 [15].

The spectrometric measurements are performed with a high-purity germanium detector. The samples are placed directly on top of the aluminum detector end cap. To calculate accurately the activities of the different produced radionuclides visible in the spectrum, a MC simulation of the radioactive decays of these different radionuclides is used to determine the absolute full energy peak (FEP) detection efficiency, a product of the detector efficiency, the solid angle, the emission probability of the radionuclide under study and the self absorption in the sample itself. The implemented detector geometry is optimized with the help of auxiliary measurements using calibrated point sources [16], by varying unknown detector properties, like e.g. the dead layer, to achieve a minimal deviation from the calibrated activities. The calculation of the signal and background of each peak is based on DIN ISO 11929:2011 [17] to calculate the net count rate. From the net count rate the activity can be determined with the FEP factor. The systematic uncertainty of this procedure is 1.7% for all radionuclides [12] and is given by the uncertainty of the FEP and the resolution of the measurement time of the read out system. The reproducibility is evaluated by the participation in internationally organized proficiency tests, e.g. by the International Atomic Energy Agency (IAEA), or by the (re-) measurement of certified sources.

The proton beam energies, proton fluences and complete measurement time for each sample is collected in Table 2. The measurement time for the targets differs. This has an influence only on the statistical uncertainty of the activation cross section for ^7Be because the half-life of ^{11}C is in the order of 20 min. In Ref. [12], the activation cross sections on carbon of a proton beam energy of 100 MeV have been published. In the results, these values are added to the table and figures, because those values have been obtained with the same setup.

2.4. Calculation of the activation cross section

For the calculation of the activity of the radionuclide under study at the end of the irradiation, A_0 , two corrections have to be considered: The decay during the approx. 40 min long transport from Essen to Dortmund and the decay during the measurement at the DLB. The activation cross section σ can then be calculated as

$$\sigma = \frac{M_T A_0}{N_A m_T \Phi_{\text{proton}} \lambda}, \quad (1)$$

where Φ_{proton} is the proton fluence, M_T and m_T are the target's molar

Table 2

Overview of experimental parameters of the processed activations and measurements during the presented experiments. The proton fluence is given with the overall uncertainty of the FC measurements which is treated as a systematic uncertainty for the activation cross section calculation. The Uncertainty of the energy is a conservative estimate of the uncertainties by the treatment machine and the energy loss of the protons in the target material.

Target	E_p [MeV]	Φ_{proton} [10^9 cm^{-2}]	Measurement time
Graphite	120 ± 1	3.80 ± 0.04	3 days
Graphite	140 ± 1	4.25 ± 0.05	3 days
Graphite	160 ± 1	4.59 ± 0.05	4 days
Graphite	180 ± 1	5.11 ± 0.06	4 days
Graphite	200 ± 1	5.49 ± 0.06	4 days
Graphite	220 ± 1	5.86 ± 0.06	9 days
Aluminum	100 ± 1	13.06 ± 0.13	20 days

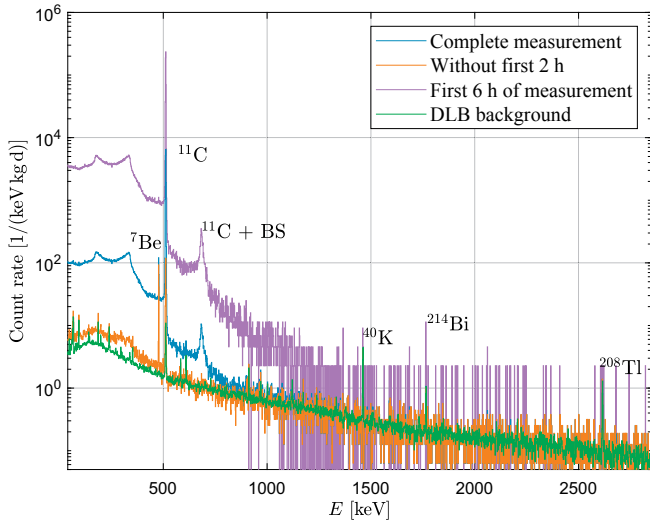


Fig. 1. Gamma-ray spectra (purple, blue and orange) of the measurement of a graphite sample irradiated with 220 MeV protons. The blue spectrum is the complete measurement (about nine days), for the orange one, the first two hours of data acquisition are neglected. The first six hours of the measurement are shown in purple. The spectra are normalized to their measurement time. The DLB background spectrum (green) is shown for comparison, several characteristic peaks are identified which are known as intrinsic contaminations of the setup, compare Ref. [11]. BS is the backscatter peak.

mass and mass, N_A is the Avogadro constant and λ is the decay constant for the radionuclide under investigation. The target area is not required for the calculation because the proton fluence is uniform across the target area.

The decay of ^{11}C during the irradiation can be neglected because the targets are irradiated within 10 s and the decay is less than 1%. All systematic uncertainties associated with the irradiation and the spectrometric measurement sum up to 2.8%, see Ref. [12] for a detailed calculation.

3. Results

3.1. Activation cross sections from carbon

Two radionuclides are visible in the gamma-ray spectra from the activated graphite, see Fig. 1. At first, the dominant annihilation peak (AP) at 511 keV, which is associated to the decay of ^{11}C , is visible within the first six hours of the measurement. About one day after the irradiation, a second gamma peak becomes visible at 478 keV, induced by the decay of ^7Be . The positron emitter ^{10}C is not visible in the AP because of the short half-life of less than one minute compared with the 40 min time of the transport.

The decay of ^{11}C is dominant during the first two hours of the measurement. A coincidence between the AP and the backscatter (BS) peak is visible at 682 keV. The BS peak is induced by photons which are scattered in the experimental setup and reach the detector afterwards. Within these first two hours of the measurement, the net count of the AP decreases by about two orders of magnitude. The other peaks observed are known to come from background processes that can be seen in the DLB background spectrum, compare Refs. [11,16].

For the AP, it is necessary to verify that no other β^+ emitting radionuclides contribute to the peak. Therefore, an exponential model is fitted to the net count rate. For all six measurements with different beam energies, the fit results are in agreement with the half-life reported in the literature, $T_{1/2}^{\text{lit}} = (20.361 \pm 0.023) \text{ min}$ [13]. The net count rate and the fit function are shown in Fig. 2 exemplary for the measurement of the sample activated with 220 MeV proton beam energy.

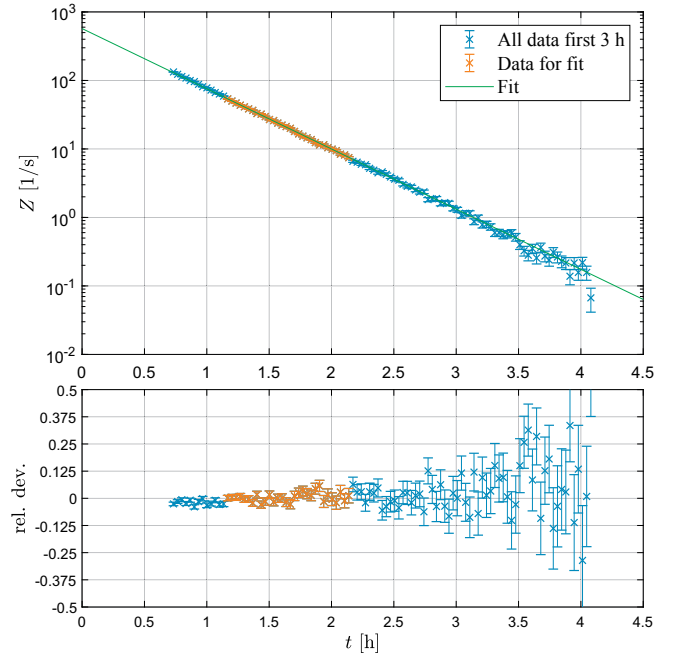


Fig. 2. Top: Net count rate Z of the AP in dependence of the time after irradiation (blue) with the fit of the decay equation (green). The fit is performed using the orange data points. Bottom: relative deviation (rel. dev.) between the data and the fit.

The first hour of data acquisition is neglected, because the dead time of the multi-channel analyzer is more than one second for each time interval and would lead to a bias in the estimated half-life. Furthermore, the net count rate of the peak is smaller than expected because of the relatively high activity of about 1 kBq. In the first hour of measurement, random coincidences of two decays are observed. At last, the data after the second hour of measurement are neglected because only one hour of the measurement is chosen for the determination of the activity from the count rate. The obtained half-life for the data shown in Fig. 2 is $(20.31 \pm 0.13) \text{ min}$.

In contrast, ^7Be decays via electron capture and emits gamma-rays with a specific energy of 478 keV. A half-life analysis is not meaningful for this radionuclide since the half-life is larger than the measurement time, compare Table 2, so a fit would result in large statistical uncertainty.

The results of the activation cross section calculation are shown in Table 3. Two different uncertainties are given in the table. The first uncertainty accounts for systematic effects, the second represents the statistical uncertainty. The target size and proton fluence are optimized to achieve low uncertainties for the ^{11}C reaction. The ^7Be peak has larger statistical uncertainty because the activation cross section is

Table 3

Activation cross sections for different proton beam energies E_p for both nuclides, ^{11}C and ^7Be . The values for $E_p = 100 \text{ MeV}$ are taken from Ref. [12]. The first uncertainty accounts for systematic effects, the second represents the statistical uncertainty. The uncertainties of the activation cross sections correspond to 1 standard deviation (s.d.).

E_p [MeV]	$\sigma(^{11}\text{C})$ [mb]	$T_{1/2}(^{11}\text{C})$ [min]	$\sigma(^7\text{Be})$ [mb]
100	$70.2 \pm 1.6 \pm 0.2$	20.45 ± 0.09	$15.9 \pm 0.4 \pm 0.6$
120	$61.5 \pm 1.4 \pm 0.3$	20.40 ± 0.10	$15.6 \pm 0.4 \pm 0.7$
140	$57.5 \pm 1.3 \pm 0.3$	20.34 ± 0.14	$14.2 \pm 0.4 \pm 0.7$
160	$53.9 \pm 1.3 \pm 0.3$	20.30 ± 0.14	$12.9 \pm 0.3 \pm 0.5$
180	$50.7 \pm 1.2 \pm 0.2$	20.30 ± 0.14	$12.2 \pm 0.3 \pm 0.5$
200	$50.2 \pm 1.2 \pm 0.2$	20.30 ± 0.14	$12.9 \pm 0.3 \pm 0.5$
220	$47.8 \pm 1.1 \pm 0.2$	20.31 ± 0.13	$11.1 \pm 0.3 \pm 0.3$

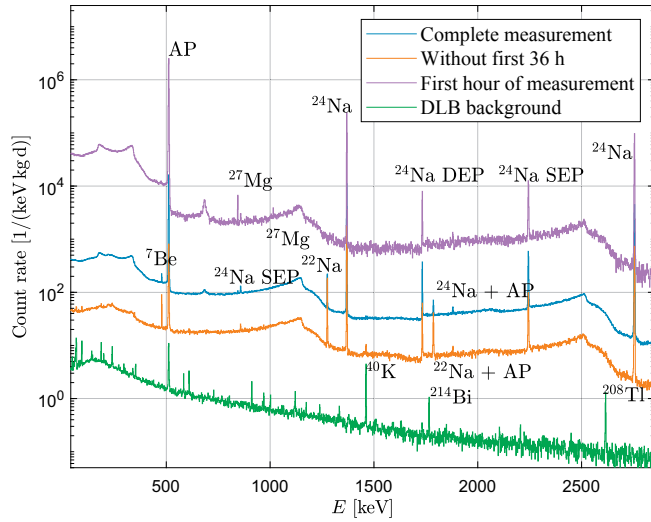


Fig. 3. Gamma-ray spectra (purple, first hour of measurement, blue, complete measurement of about three weeks and orange, without the first 36 h of measurement) of the measurement of the aluminum sample activated with 100 MeV protons from different time intervals of data taking. The spectra are normalized to their measurement time. SEP and DEP are the single escape peak and double escape peaks. The DLB background spectrum (green) is shown for comparison, several characteristic peaks are identified.

lower by a factor of four. The activity of ^7Be is also smaller because of the longer half-life of about 53 days [13].

3.2. Reproducibility test with aluminum

The gamma-ray spectrum of the measurement of the irradiated aluminum sample is shown in Fig. 3. Four radionuclides are identified, namely ^7Be , ^{22}Na , ^{24}Na and ^{27}Mg . In Table 4, the activation cross sections from our previous work, Ref. [12], are compared to the current results. In addition, the radionuclide ^{27}Mg can be identified with two peaks at 844 keV and 1014 keV and a half-life of (9.458 ± 0.012) min within the first hour of the measurement. The uncertainty of this result is about 10%, and dominated by the dead time of the multi-channel analyzer, which is more than one second per data acquisition interval of 2 min within the first hour. For the three radionuclides, ^7Be , ^{22}Na and ^{24}Na , deviations of the order of one percent are measured. The activation of the aluminum sample with a 100 MeV proton beam results in comparable values as the one already published in Ref. [12].

4. Discussion

4.1. The $^{nat}\text{C}(p,x)^{11}\text{C}$ -excitation function

The activation cross section as a function of the proton beam energy, the excitation function, is shown in Fig. 4, together with the data

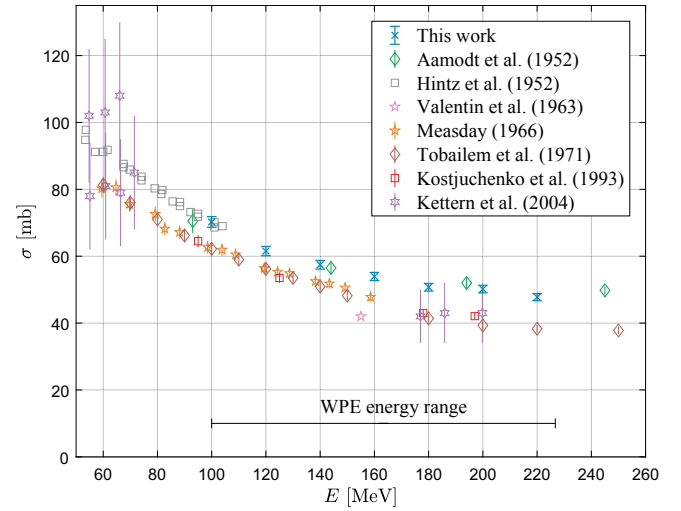


Fig. 4. Activation cross section data of the $^{nat}\text{C}(p,x)^{11}\text{C}$ -excitation function with the data from the EXFOR database [3–9]. “This work” includes the activation cross section for 100 MeV from Ref. [12].

taken from the EXFOR database. The measurements cluster around two different excitation functions which are about 10% apart, independent of the proton beam energy. Our measurements are in agreement with the data from Aamodt et al. [3] and Hintz et al. [4]. In contrast, measurements by Toballem et al. [7], Measday [6] and Kostjuchenko et al. [8] yield systematically lower activation cross sections. For the calculation of absolute cross sections, Refs. [6,8] both describe a relative cross section measurement with a normalization to a reference cross section at a proton beam energy of 98.1 MeV. However, Ref. [7] presents comparable absolute cross sections. A normalization of the results is not described in Ref. [7]. Kettern et al. [9] have obtained cross sections with larger uncertainties which agree with all data collected in the EXFOR database.

The uncertainties of the activation cross sections of this work are smaller than the uncertainties of the ones from the EXFOR database in the used energy range. The latter, if given at all, range between 5% and 20% for proton beam energies between 100 and 220 MeV.

There are many possible reasons for the observed difference of the two excitation functions. Different half-life estimates for ^{11}C have been used in the analysis of ^7Be in the past, resulting a large discrepancy. Another reason is the method of relative measurements with the risk of normalization on a systematic to large or small cross section. Furthermore, different measurement technique have been used to determine activation cross sections. The presented method uses one thin single target per irradiation. In the past, the foil-stack technique has been used in different prior studies. Additionally, the methodology of determination of the number of protons differs for the measurements of absolute activation cross sections.

4.2. The $^{nat}\text{C}(p,x)^7\text{Be}$ -excitation function

Since there are two different excitation functions for the reaction to ^{11}C , a systematic overestimation of these cross section might be concluded from the data. The advantage of the gamma-ray spectrometry is that the residual ^7Be can be evaluated as well. A systematic error in the experimental method would have the same influence on both reactions, so this reaction can be used as a monitoring for experimental errors. However, the measurement of the $^{nat}\text{C}(p,x)^7\text{Be}$ -reaction has no relevance for proton beam range verification. Furthermore, both reactions describe the fragmentation of carbon ions on hydrogen, e.g. for carbon ion therapy.

The measured activation cross sections for the $^{nat}\text{C}(p,x)^7\text{Be}$ -reaction are shown in Fig. 5 together with the corresponding data from the

Table 4

Comparison of the activation cross section for ^7Be , ^{22}Na and ^{24}Na from aluminum at 100 MeV proton beam energy to demonstrate the reproducibility. The uncertainty of the deviation (dev.) is an overall uncertainty, accounting systematic and statistical effects by square addition. In addition the activation cross section for ^{27}Mg is listed. This result is not available (n.a.) from Ref. [12]. The uncertainties correspond to 1 s.d.

Nuclide	$\sigma_{06/2018}$ [mb]	$\sigma_{05/2017}$ [mb] [12]	dev. [%]
^7Be	$1.11 \pm 0.02 \pm 0.03$	$1.13 \pm 0.02 \pm 0.04$	2 ± 5
^{22}Na	$19.7 \pm 0.3 \pm 0.3$	$19.9 \pm 0.3 \pm 0.3$	1 ± 4
^{24}Na	$11.58 \pm 0.16 \pm 0.03$	$11.48 \pm 0.15 \pm 0.04$	-1 ± 3
^{27}Mg	$0.13 \pm 0.01 \pm 0.01$	n.a.	n.a.

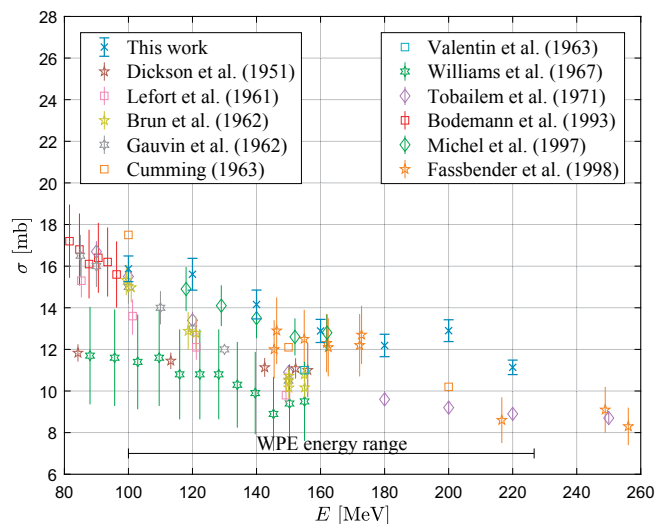


Fig. 5. Activation cross section data of the $^{nat}\text{C}(\text{p},\text{x})^7\text{Be}$ -excitation function with the data from the EXFOR database [5,7,18–26]. “This work” includes the activation cross section for 100 MeV from Ref. [12].

EXFOR database [10]. Most groups have used the $^{nat}\text{C}(\text{p},\text{x})^{11}\text{C}$ -reaction as a monitor reaction to calculate the cross section for this reaction indirectly. The cross sections obtained by Dickson et al. [18] and Williams et al. [23] are up to 25% smaller than the ones measured by several other groups. In Ref. [18], a half-life of 20.0 min is used for the decay of ^{11}C to correct the measured activity for 5 h of decay time. In 2019, $T_{1/2}^{11\text{C}} = (20.361 \pm 0.023)\text{min}$ is considered to be the correct half-life of ^{11}C [13]. This consequently causes a difference of 20% for the activation cross sections when used during the calculation. A larger half-life would lead to larger activation cross section, and the results would agree with the current results.

As described for the production of ^7Be in Ref. [12], the result for the proton beam energy of 100 MeV is in agreement with several measurements, e.g. those from Brun et al. [20], Gauvin et al. [21], Tobialem et al. [7] and Bodemann et al. [24]. These groups measured a larger dependence on the proton beam energies compared to the measurements presented in this work. In contrast to the $^{nat}\text{C}(\text{p},\text{x})^{11}\text{C}$ -reaction, where the obtained cross sections of this work are 10% larger, the $^{nat}\text{C}(\text{p},\text{x})^7\text{Be}$ -reaction result agrees with Ref. [7] at 100 MeV. A systematic overestimation, which might be derived from the results for the production of ^{11}C , can be rejected based on this observation. Only in Ref. [7], both reactions have been measured directly in the past. A systematic overestimation of nuclear activation cross sections with the chosen experimental setup would yield in larger results for all obtained nuclear reactions. For the reaction from natural carbon to ^7Be and for the activation cross sections from aluminum, Ref. [12], both the data from the EXFOR database are reproduced with the setup.

For proton beam energies up to about 180 MeV, the results agree with the ones from Michel et al. [25] and Fassbender et al. [26]. However, the change in the activation cross sections between 180 MeV and 220 MeV, which is visible in Ref. [26], is not reproduced in this work. Furthermore, at 200 MeV and 220 MeV, the results are larger than the data from the EXFOR database [10].

The $^{nat}\text{C}(\text{p},\text{x})^7\text{Be}$ -activation cross sections are measured directly and independently of the $^{nat}\text{C}(\text{p},\text{x})^{11}\text{C}$ -activation cross sections. There are larger uncertainties for the $^{nat}\text{C}(\text{p},\text{x})^7\text{Be}$ -activation cross sections compared with the other reaction, which is caused by the activity of ^7Be as discussed in Section 3. The larger activation cross sections at 200 MeV compared with the beam energies of 180 MeV and 220 MeV is given by the larger statistical uncertainty. Furthermore, larger activation cross sections are not observed for the production of ^{11}C .

4.3. Activation of aluminum

The results for the activation cross sections on aluminum show, that a systematic drift of experimental parameters like the number of protons or the detector efficiency of the DLB is not observed within this one year. This is expected, since the WPE is clinical proton beam therapy facility and the DLB participates several proficiency tests over time. Concluding, all measured absolute activation cross sections refer to the same experimental conditions. Furthermore, the results of Ref. [12] are in agreement with several data collected in the EXFOR database and IAEA recommendations for monitor reaction. Since there is no significant difference between those results and the current one, the results presented in this study are in agreement with the data from the literature of activation cross sections from aluminum.

5. Conclusion

In summary, this work presents an accurate measurement of activation cross sections of protons on graphite targets with systematic uncertainties of the order of 2.8%. The statistical uncertainties are nearly negligible for the activation cross sections for the production of ^{11}C . For ^7Be , the statistical uncertainties range in the same order of magnitude compared with the systematic uncertainties. The different radionuclides are identified and quantified with low-level gamma-ray spectrometry. The results are compared with data from the EXFOR database on a detailed level to show differences and similarities of the procedure to estimate the activation cross sections.

While a detailed discussion of the experimental setup and its uncertainties can be found in Ref. [12], this work presents the measurement of the activation cross sections in steps of 20 MeV. The results from the EXFOR database are reproduced in this experimental study for ^7Be . For ^{11}C , the presented results are in agreement with the upper excitation function, which can be derived from the data in the EXFOR database. Furthermore, lower uncertainties for the activation cross sections in the energy range from 100 MeV to 220 MeV are achieved with the experimental procedure. Only Ref. [27] achieved a lower uncertainty at the 0.5%-level but measured the cross section only at one single proton beam energy at 98.1 MeV.

The reproducibility of the experimental setup is shown with the irradiation and subsequent activation measurement of a second aluminum sample. The deviations of the results are lower than the overall uncertainty of the measurements. The prior results from Ref. [12] agree with the data from the EXFOR database as well as with the new results.

Declaration of Competing Interest

None.

Acknowledgments

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